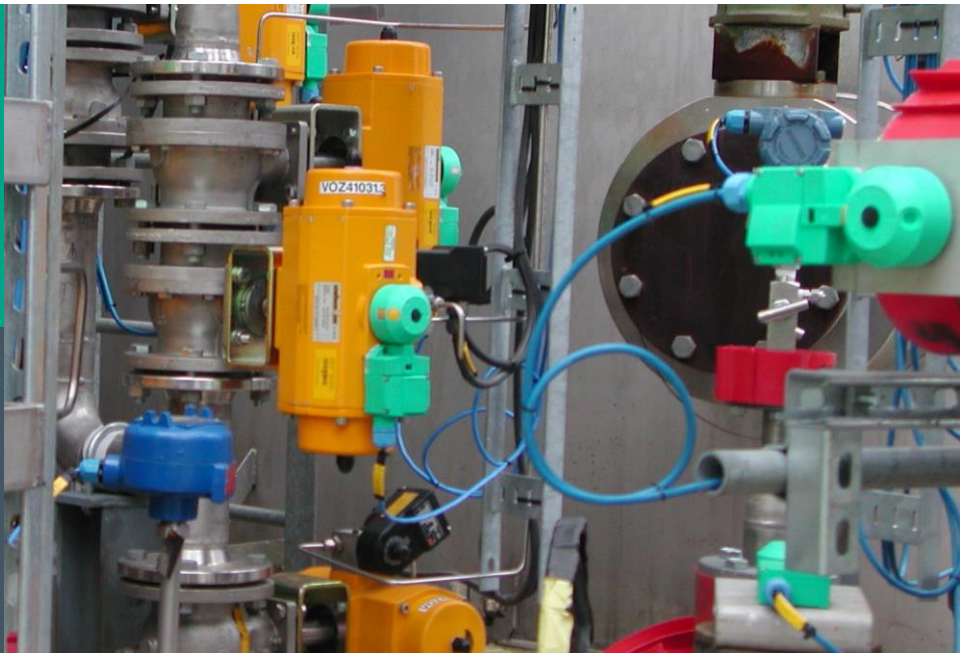
Explosion Protection

Electrical Equipment for Hazardous Areas



Your automation, our passion.

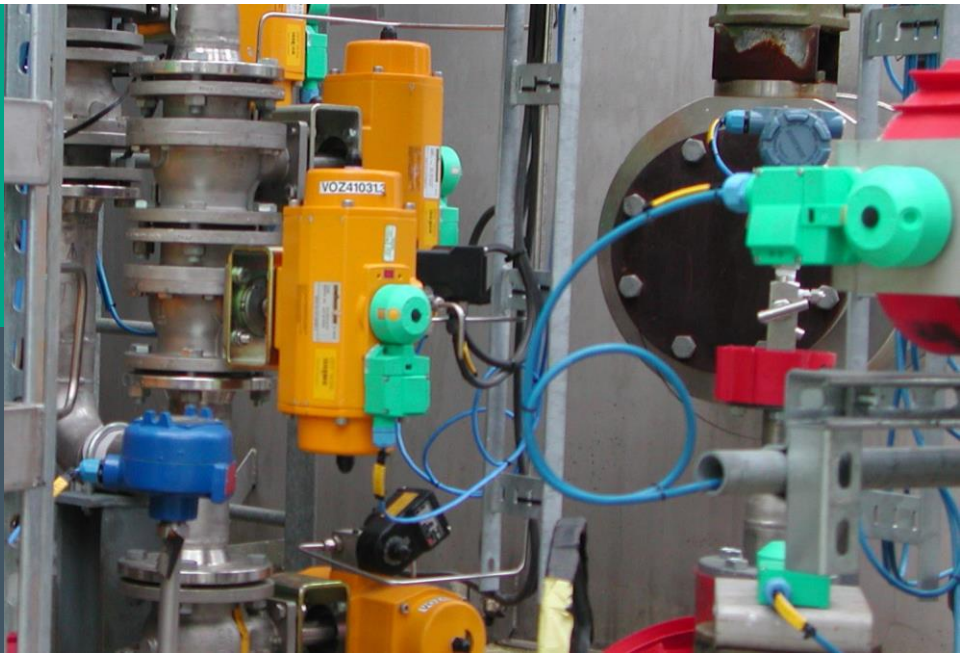
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Overview

1. **General Requirements**
   Absence of Ignition Sources, Technical Requirements, Documentation, Marking
2. **Present Types of Protection – Gas**
   Functional Principle and Marking “i”, “d”, “e” and “p”, Examples
3. **Present Types of Protection – Dust**
   Functional Principle and Marking of “t”, “i”, “m” and “p”, Examples
4. **Previous Marking**
   Examples

1. General Requirements

Electrical Equipment for Hazardous Areas



Equipment for the Use in Hazardous Areas



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Manufacturers of equipment for the use in hazardous areas have to:

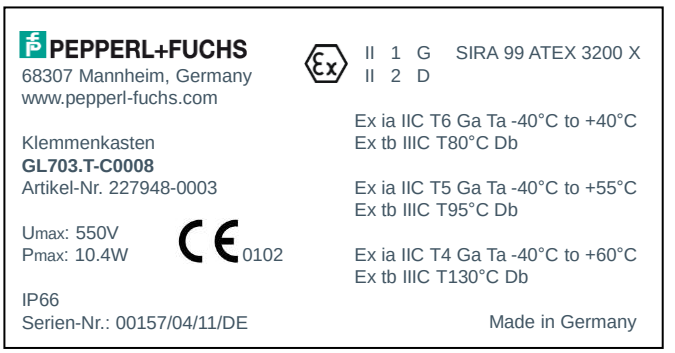
- Describe the intended use (e.g. zone, ambient temperature, operation voltage etc.)
- Make sure that there is no ignition source (even under fault conditions – where required)
- Attach a marking and issue a documentation
  - → *Series of standards EN 60079-… comprises present requirements*
  - → *Covers both mining/non mining equipment as well as gas/dust equipment*

Demands on Explosion Proof Equipment

Depending on the specific type of protection the following requirements, amongst others, have to be met:

- Classification in groups (I, II or III)
- Temperature behaviour (max. surface temperature, ambient temperature)
- Mechanical stability (impact resistance etc.)
- Demands on gaskets (IP rating)
- Metallic materials (light metals)
- Non metallic materials (e.g. electrostatic charge, UV resistance)
- Cable entries
- Clearances and creepage distances, equipotential bonding
- Additional demands on specific pieces of equipment (e.g. motors, luminaires)
- Type tests
- Documentation
- Marking

Example: Junction Box Label



Specific Ex Marking – Gas Equipment

Ex ia IIC T6 Ga T_a = - 40°C … + 40°C
❶ ❷ ❸ ❹ ❺ ❻

	Meaning	Note
❶	Explosion protected	
❷	Type(s) of protection	E.g. intrinsic safety “i”, flameproof enclosure “d” etc.
❸	Group	II: non mining, gas, subclassification A, B or C
❹	Surface temperature	In case of gas equipment normally a temperature class
❺	Equipment protection level	„Very high“, „high“, „enhanced“
❻	Ambient temperature	E.g. in case of deviations from the standard range

Categories vs. Equipment Protection Levels



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The traditional types of have been suitable for gas areas only:

- Meanwhile some types of protection are also suitable for dust areas
- Additionally there are different safety requirements for the types of protection depending on the intended use in zone 0, 1, 2 as well as 20, 21, 22
  - → *The marking has to permit a suitable differentiation*

Zone, Category, EPL – Gases and Vapours

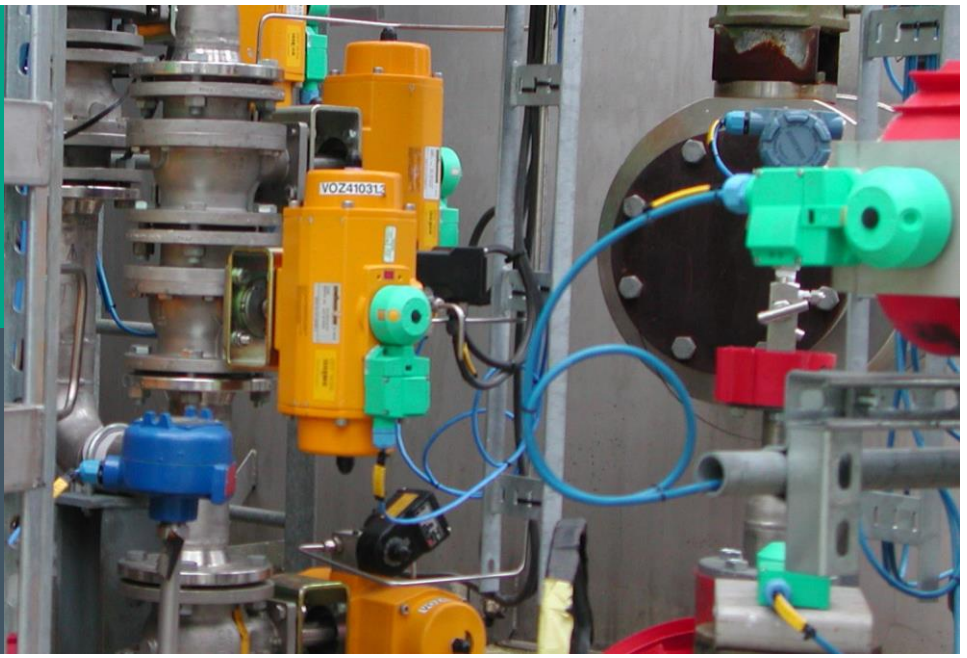


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IEC / ATEX	IEC		ATEX	
Zone	Suitable group	Suitable EPL	Suitable group	Suitable category
0	II (A, B, C)	Ga	II	1G
1		Ga or Gb		1G or 2G
2		Ga, Gb or Gc		1G, 2G or 3G

2. Present Types of Protection – Gas

Functional Principle and Marking “i”, “d”, “e” and “p”, Examples



Intrinsic Safety – “i”



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Functional principle of intrinsic safety:

- Circuits are operated at limited voltage, current and power
- Energy storages (inductances, capacitances) are limited as well
  - → *Generated sparks do not exceed the minimum ignition energy of the gas*

Protective Measures – “i”



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Applied protective measures:

- A barrier (“associated apparatus”) limits voltage, current and power
- Field devices and cables have defined (limited) energy storages
- Protection from the intrusion of energy

Term: Intrinsic safety “i”

Type of protection based upon the restriction of electrical energy within the apparatus and of interconnecting wiring exposed to an explosive atmosphere to a level below that which can cause ignition by either sparking or heating effects.

EN 60079-14:2014; 3.5.1

Term

Citation

Literature

Marking – “i”

Ex ia IIC T6 Ga

Ex ib IIC T6 Gb

Ex ic IIC T6 Gc

- Type of protection “i“: intrinsic safety
- “a”, “b”, “c”: level of protection (electrical circuit only!)
- Group “II”: non mining equipment, gas
- Subclassification “A”, “B”, “C”: spark energy
- Marking “T*”: maximum surface temperature
- Equipment protection level “G*”: degree of likelihood to become an ignition source



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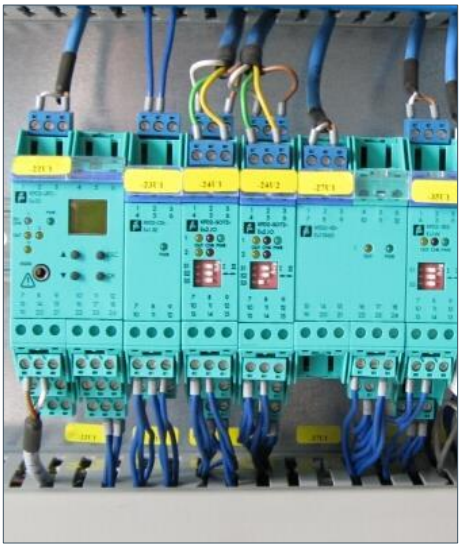
Marking – “i”

[Ex ia Ga] IIC

[Ex ib Gb] IIC

[Ex ic Gc] IIC

- Marking „[ ]“: Installation in the non-hazardous area
- No surface temperature or temperature class marking



Peter Leitenmaier, TÜV SÜD Industrie Service

Particular Demands – “i”



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Reasonable care should be taken with respect to:

- The verification of intrinsic safety
- The separation to other circuits
  - → *Chapter „Intrinsic Safety“*

Flameproof Enclosure – “d”



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Functional principle of flameproof enclosure:

- Effective sources of ignition are located inside an enclosure
- Ingress of explosive atmosphere is possible
- Internal explosions do not destroy the enclosure
  - → *Prevention of an explosion propagation to the explosive atmosphere surrounding the enclosure*

Protective Measures – “d”



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Applied Protective Measures:

- Application of robust enclosures which can withstand the internal explosion pressure
- Compliance with specific joint gaps depending on the required explosion group IIA, IIB or IIC

Term: Flameproof Enclosure “d”

Type of protection in which the parts capable of igniting an explosive gas atmosphere are provided with an enclosure which can withstand the pressure developed during an internal explosion of an explosive mixture and which prevents the transmission of the explosion to the explosive gas atmosphere surrounding the enclosure.

EN 60079-14:2014; 3.3.1

Term

Citation

Literature

Marking – “d”

Ex da IIC T4 Ga

Ex db IIC T4 Gb

Ex dc IIC T4 Gc

- Marking „d“: flameproof enclosure
- Sub-classification “A”, “B”, “C”: gap dimensions



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Particular Demands – “d”



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Reasonable care should be taken with respect to:

- Selection of appropriate cable glands and blanking elements
- Selection of appropriate cable
- Protection of flameproof joints
- Alteration of the internal components

Increased Safety – “e”



Flowserve

Functional principle of increased safety:

- Potential (but not effective!) sources of ignition are located in an enclosure
- Ingress of explosive atmosphere is possible
  - → *Prevention of potential sources of ignition to become effective*

Protective Measures – “e”



Flowserve

Applied Protective Measures:

- Reinforced insulation (e.g. motor winding)
- Protection against self-loosening (e.g. terminals)
- Overload protection (e.g. thermal monitoring)
- Increased IP-rating and mechanical stability (e.g. junction box, cable gland)

Term: Increased Safety “e”

Type of protection applied to electrical equipment in which additional measures are applied so as to give increased security against the possibility of excessive temperatures and of the occurrence of arcs and sparks in normal service or under specified abnormal conditions.

EN 60079-14:2014; 3.4.1

Term

Citation

Literature

Marking – “e”

Ex eb IIC T3 Gb

Ex ec IIC T3 Gc

- Marking „e“: increased safety
  - ➝ *Why is there a marking IIC?*



Flowserve

Particular Demands – “e”



Flowserve

Reasonable care should be taken with respect to:

- Selection of appropriate cable glands, blanking elements, terminals
- Protection of motors against overload
- Operation of motors with frequency inverters
- Modification of junction boxes

Pressurisation – “p”



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Functional principle of pressurisation:

- Sources of ignition are located inside an enclosure
- Enclosure is filled with a protective gas
- Ingress of explosive atmosphere is prevented
  - → *Separation of explosive atmosphere and source of ignition*

Protective Measures – “p”



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Applied Protective Measures:

- Enclosure is under pressure
- Consideration of purging procedures

Term: Pressurisation “p”

Technique of guarding against the ingress of the external atmosphere into an enclosure by maintaining a protective gas therein at a pressure above that of the external atmosphere.

EN 60079-14:2014; 3.7.1

Term

Citation

Literature

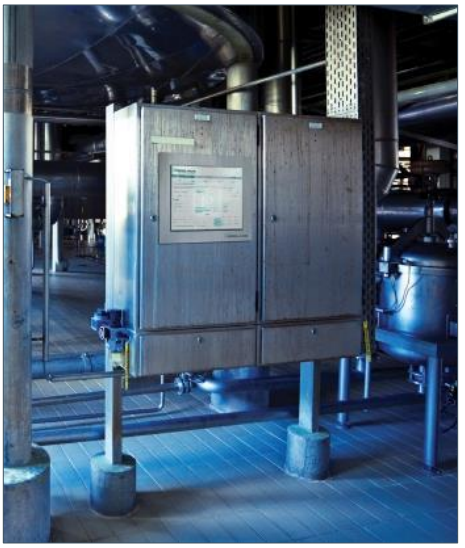
Marking – “p”

Ex pxb IIC T3 Gb

Ex pyb IIC T3 Gb

Ex pzc IIC T3 Gc

- Marking „p“: pressurisation
  - → *Why is there a marking x, y or z?*



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Protective Measures – “p”



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Reasonable care should be taken with respect to:

- The combination of uncertified enclosures with certified control devices is not admissible
- Consideration of purge times
- Installation of spark and particle barriers as required
- Alarm and switch-off depending on EPL and internal equipment

Particular Demands – “p”



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Marking	External zone	Internal zone
pxb	1	-
pyb	1	2
pzc	2	-

Optical Radiation – ”op”



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Functional principle:

- Limitation of the radiant energy or
- Separation of ignition capable radiation and explosive atmosphere

Marking – “op”

Ex op is IIC T6 Ga

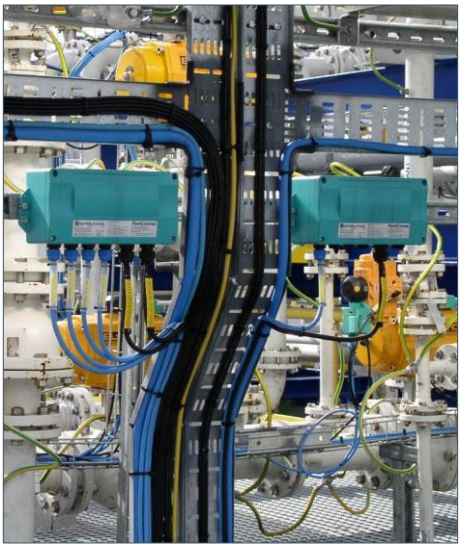
- Equipment for zone 0 („Ga“)
- Limitation of the radiant energy („IIC“)

Ex op pr II T4 Gb

- Protection of the transmitting medium from damage

Ex op sh II T4 Gc

- Switch-off in case of escape of the radiation from the transmitting medium



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Marking “m”

Ex ma IIC T4 Ga

Ex mb IIC T4 Gb

Ex mc IIC T4 Gc

- Marking „m“: encapsulation



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Marking “o”

Ex ob IIC T3 Gb

Ex oc IIC T3 Gc

- Marking “o”: oil immersion

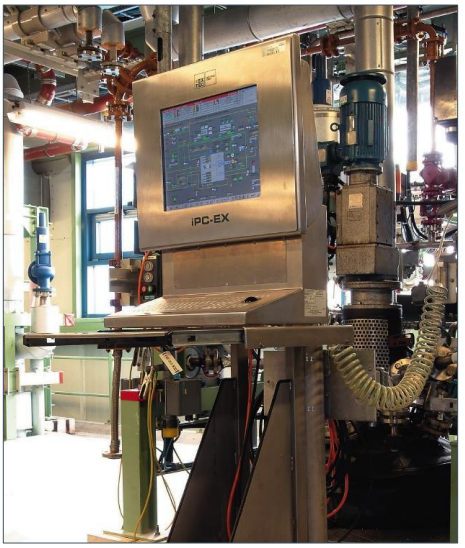


Oliver Frühschutz

Marking “q”

Ex qb IIC T4 Gb

- Marking “q”: powder filling



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Marking “n”

Ex nA IIC T3 Gc

Ex nC IIC T3 Gc

Ex nR IIC T3 Gc

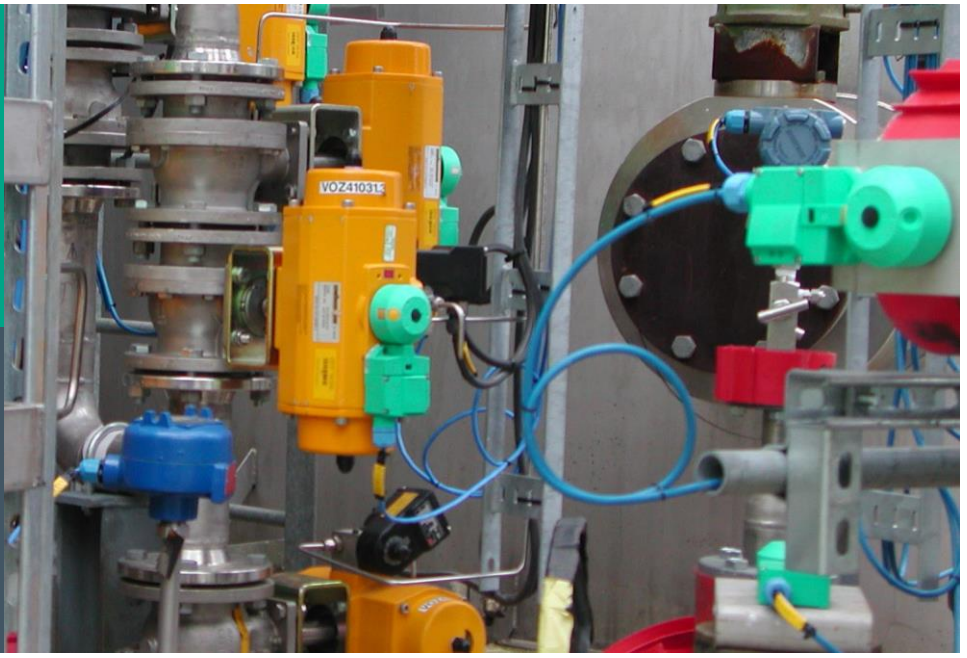
- Marking „nA“: non sparking (comparable to “ec”)
- Marking „nC“: protected sparking (comparable to “mc” or “dc”)
- Marking „nR“: restricted breathing enclosure (limited ingress of gas or vapour)



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3. Present Types of Protection – Dust

Functional Principle and Marking of “t”, “i”, “m” and “p”, Examples



Present Types of Protection - Dust



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Currently the following types of protection exist:

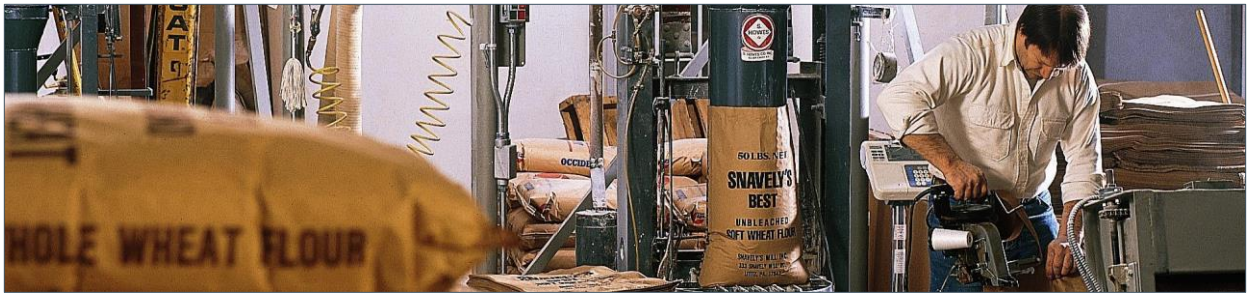
- Intrinsic safety “i”
- Encapsulation “m”
- Pressurisation “p”
- Protection by enclosure “t”

Specific Ex Marking – Dust Equipment

Ex ❶ ia ❷ IIIC ❸ T80°C ❹ Da ❺ T_a = - 40°C … + 40°C ❻

	Meaning	Note
❶	Explosion protected	
❷	Type(s) of protection	E.g. intrinsic safety “i”, protection by enclosure “t” etc.
❸	Group	III: non mining, dust, subclassification A, B or C
❹	Surface temperature	Actual surface temperature [°C]
❺	Equipment protection level	„Very high“, „high“, „enhanced“
❻	Ambient temperature	E.g. in case of deviations from the standard range

Zone, Category, EPL – Gases and Vapours



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IEC / ATEX	IEC		ATEX	
Zone	Suitable group	Suitable EPL	Suitable group	Suitable category
20	III (A, B, C)	Da	II	1D
21		Da or Db		1D or 2D
22		Da, Db or Dc		1D, 2D or 3D

Protection by Enclosure – “t”



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Functional principle of protection by enclosure:

- Prevention or restriction of dust ingress into the enclosure
  - → *Separation of explosive dust atmosphere from sources of ignition*

Protective Measures – “t”



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Applied protective measures:

- Application of robust enclosures (impact test etc.)
- Ensuring a sufficient IP rating

Term: Protection by Enclosure “t”

Type of protection whereby all electrical equipment is protected by an enclosure to avoid ignition of a dust layer or cloud.

EN 60079-14:2014; 3.12

Term

Citation

Literature

Marking – “t”

Ex ta IIIC T80°C Da

Ex tb IIIC T80°C Db

Ex tc IIIC T80°C Dc

- Marking „t“: protection by enclosure
- Group III: non mining equipment, dust
- Sub-classification “A”, “B”, “C”: characteristics of the dust (particle size, conductivity)



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Particular Demands – “t”

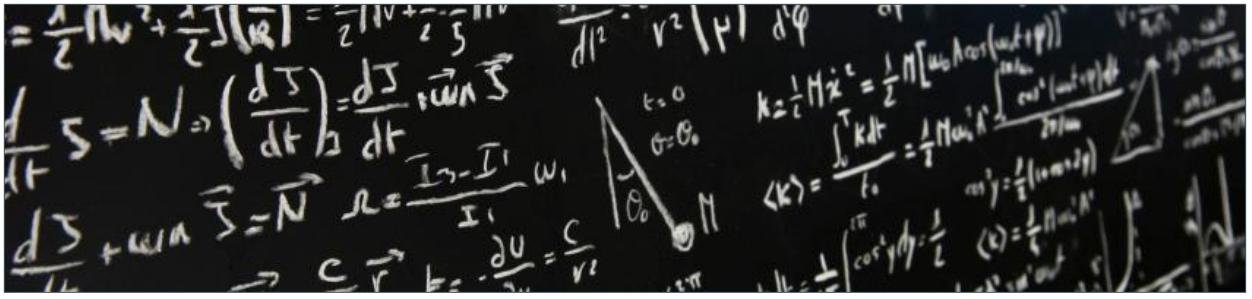


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Reasonable care should be taken with respect to:

- Selection of appropriate cable glands, blanking elements
- IP rating and type of combustible dusts IIIA, IIIB, IIIC
- Consideration of a safety margin between surface and ignition temperatures

Ignition Temperatures and Safety Margins



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Required safety margins between surface temperature T_{max} and ignition temperatures:

- T_{max} must not exceed 2/3 of the ignition temperature of the dust cloud (T_{CL})
- T_{max} must be at least 75 K below the ignition temperature of the dust layer (T_{5mm})
  - → *To be considered by the designer of the system*

Example: Power Supply in Zone 21

Ex tb IIIC T110°C Db, cellulose application:

- T_{CL} = 540 °C
- T_{5mm} = 350 °C
  - → *Values e.g. from material safety data sheet (MSDS)*
- T_{max} = 2/3 T_{CL} = 540 °C · 2/3 = 360 °C
- T_{max} = T_{5mm} – 75 K = 350 °C - 75 K = 275 °C
  - → *Maximum surface temperature of the luminaire: 110 °C, i.e. application is admissible*
  - → *Otherwise: prevention of the formation of dust layers (e.g. by a protective cover or cleaning instructions)*



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Ex t: Level of Protection, Group, IP-Rating

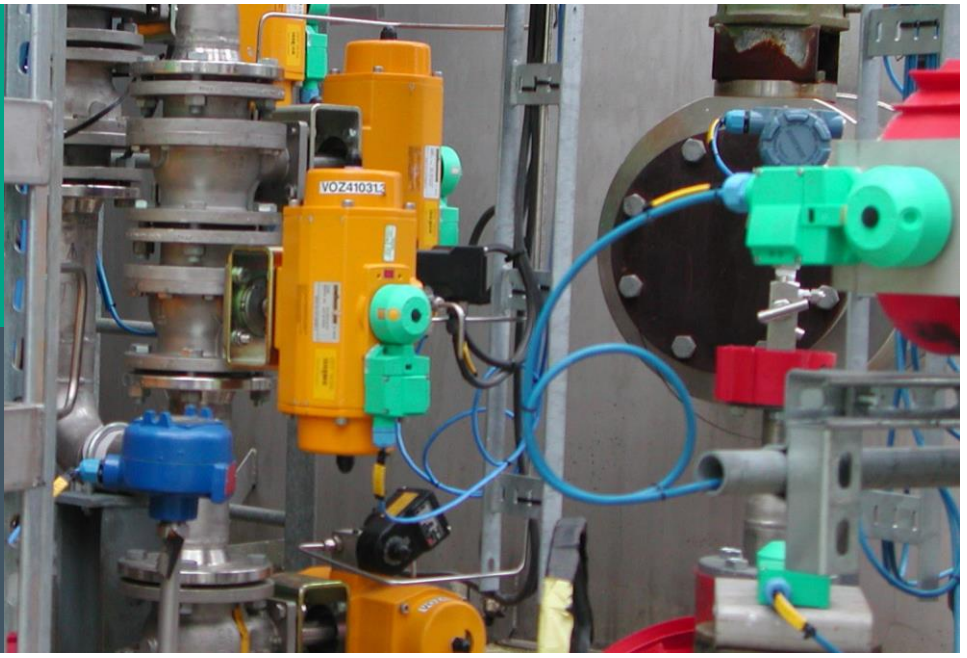
Level of Protection	Typical Zone	IIIC	IIIB	IIIA
ta	20	IP 6X	IP 6X	IP 6X
tb	21	IP 6X	IP 6X	IP 5X
tc	22	IP 6X	IP 5X	IP 5X

Classification of combustible dusts:

- IIIC: conductive dusts
- IIIB: non-conductive dusts
- IIIA: fibres or flyings

4. Previous Marking

Examples



Previous and Present Marking

Previous marking	Present marking	Note
EEx ia IIC T6	Ex ia IIC T6 Ga	
EEx ib IIC T6	Ex ib IIC T6 Gb	
EEx nL IIC T6	Ex ic IIC T6 Gc	“nL” has been replaced by “ic”
[EEx ia] IIC	[Ex ia Ga] IIC	
[EEx ib] IIC	[Ex ib Gb] IIC	
[EEx nL] IIC	[Ex ic Gc] IIC	“nL” has been replaced by “ic”
-	Ex da IIC T4 Ga	
EEx d IIC T4	Ex db IIC T4 Gb	
-	Ex dc IIC T4 Gc	Partly comparable to “nC”
EEx e II T4	Ex eb IIC T4 Gb	
-	Ex ec IIC T4 Gc	Comparable to “nA”

Previous and Present Marking

Previous marking	Present marking	Note
EEx p II T4	Ex pxb IIC T4 Gb	
-	Ex pyb IIC T4 Gb	
-	Ex pzc IIC T4 Gc	
-	Ex ma IIC T4 Ga	
EEx m II T4	Ex mb IIC T4 Gb	
-	Ex mc IIC T4 Gc	Partly comparable to “nC”
EEx o II T3	Ex ob IIC T3 Gb	
-	Ex oc IIC T3 Gc	
EEx q II T3	Ex q IIC T3 Gc	

Previous and Present Marking

Previous marking	Present marking	Note
EEx iaD 20 IP67 T80°C	Ex ia IIIC T80°C Da	
EEx ibD 21 IP67 T80°C	Ex ib IIIC T80°C Db	
-	Ex ic IIIC T80°C Dc	
EEx pD 20 IP66 T90°C	Ex p IIIC T90°C Db	
-	Ex ma IIIC T80°C Da	
EEx mD 21 IP67 T80°C	Ex mb IIIC T80°C Db	
-	Ex mc IIIC T80°C Dc	
Ex tD A20 IP67 T90°C	Ex ta IIIC T80°C Da	
Ex tD A21 IP67 T90°C	Ex tb IIIC T80°C Db	
Ex tD A22 IP67 T90°C	Ex tc IIIC T80°C Dc	